

Supporting Information

for

Signal or Redundancy? The Incremental Value of Technical Indicators for Volatility Forecasting

Tables S1-S38

Supporting Information Contents

SI-A. Data, Sample Construction and Data-Quality Audit (S1-S5)

SI-B. Variable Definitions, Feature Construction and Technical-Indicator Formulas (S6-S9)

SI-C. Forecasting Protocol, Model Specifications, Hyperparameter Selection and Refit Design (S10-S16)

SI-D. Extended Forecast Results and Robustness Tables (S17-S22C)

SI-E. Technical-Indicator Family Ablations (S23-S26)

SI-F. Liquidity-Conditioned Technical-Indicator Value and Formal State Contrasts (S27-S30)

SI-G. GJR-GARCH Benchmark, Fit Diagnostics and Forecast-Positivity Audits (S31-S38)

Supporting Information A

Data, Sample Construction and Data-Quality Audit

A.1 Raw data universe and chronology

The master NSE panel contains 107,930 stock-date observations for 52 listed firms observed over 2,488 market dates from 1 August 2016 to 31 July 2026. The development period ends on 29 December 2023 and contains 1,847 market dates. The locked pseudo-out-of-sample period begins on 2 January 2024 and ends on 31 July 2026. The study determines sample eligibility using development-period information only and excludes all 2024–2026 observations from decisions about whether a firm enters the main forecasting sample.

Table S1. Overall preprocessing audit

Audit item	Count	Interpretation
Raw stock-date rows	107,930	Master panel before eligibility filtering
Raw firms	52	Full listed-firm universe in the master file
Market dates	2,488	2016-08-01 to 2026-07-31
Development market dates	1,847	Used for eligibility and model development
Eligible firms	29	Main stock-level forecasting sample
Eligible sectors	9	Sector classification retained for description/heterogeneity
Invalid OHLC rows	135	0.125% of raw rows; not repaired for range-based construction
Missing volume rows	977	0.905% of raw rows; treated as missing
Numeric zero-volume rows	876	0.812% of raw rows; retained as genuine zeros
Eligible-sample corporate-action events	14	Events triggering segment resets

A.2 Development-only firm eligibility

A firm is eligible for the main sample if two conditions are satisfied during the development period: (i) it is observed on at least 90% of the 1,847 development market dates, and (ii) no more than 10% of its development-period volume observations are missing. The study fixes these rules before opening the locked test sample. They ensure sufficient continuity for multi-day OHLC volatility targets and rolling technical indicators without conditioning on future forecast performance.

The coverage criterion is the binding sample restriction in the present data. All 52 firms have a development-period missing-volume rate below the 10% threshold in the eligibility audit. Twenty-nine firms meet the 90% market-date coverage rule, and 23 do not. The least-covered retained firm is Sasini with 90.15% development coverage, whereas the most-covered excluded firm is Sameer Africa Ltd with 88.79%. The median development coverage among retained firms is 99.73%.

Table S2. Complete 52-firm development-period eligibility audit

Firm	Sector	Observed dev. dates	Coverage (%)	Missing volume (%)	Eligible
CIC Insurance Group	Insurance	1847	100.00	0.00	Yes
Co-operative Bank (Kenya)	Banking	1847	100.00	0.00	Yes
KCB Bank	Banking	1847	100.00	0.00	Yes
KenGen	Energy and Petroleum	1847	100.00	0.00	Yes
Kenya Re	Insurance	1847	100.00	0.00	Yes
Safaricom PLC	Telecommunication and Technology	1847	100.00	0.00	Yes
Britam Holdings	Insurance	1846	99.95	0.00	Yes
Centum	Investment	1846	99.95	0.00	Yes
Equity Group	Banking	1846	99.95	0.00	Yes
KPLC	Energy and Petroleum	1846	99.95	0.00	Yes

Firm	Sector	Observed dev. dates	Coverage (%)	Missing volume (%)	Eligible
Absa Bank Kenya	Banking	1845	99.89	0.00	Yes
East African Breweries	Manufacturing and Allied	1843	99.78	0.00	Yes
NCBA Bank	Banking	1843	99.78	0.00	Yes
Nairobi Business Ventures	Commercial and services	1843	99.78	0.00	Yes
Nation Media Group	Commercial and services	1842	99.73	0.00	Yes
Home Africa	Investment	1841	99.68	0.00	Yes
JUBILEE HOLDINGS	Insurance	1840	99.62	0.00	Yes
Nairobi Securities Exchange	Investment Services	1836	99.40	0.00	Yes
Standard Chartered Bank	Banking	1836	99.40	0.00	Yes
HF Group	Banking	1835	99.35	0.00	Yes
Eveready	Commercial and services	1824	98.75	0.00	Yes
Uchumi Supermarkets	Commercial and services	1818	98.43	0.00	Yes
WWP Scangroup	Commercial and services	1809	97.94	0.00	Yes
Carbacid Investments Ltd	Manufacturing and Allied	1764	95.51	0.00	Yes
Longhorn Publishers	Commercial and services	1726	93.45	0.00	Yes
Total Energy	Energy and Petroleum	1720	93.12	0.00	Yes
Diamond Trust Bank	Banking	1717	92.96	0.00	Yes
Flame Tree Group	Manufacturing and Allied	1686	91.28	0.00	Yes
Sasini	Agriculture	1665	90.15	0.00	Yes
Sameer Africa Ltd	Commercial and services	1640	88.79	0.00	No
Liberty Kenya Holdings	Insurance	1625	87.98	0.00	No
I&M Bank	Banking	1586	85.87	0.00	No
Stanbic Holdings	Banking	1574	85.22	0.00	No
BAT Kenya	Manufacturing and Allied	1502	81.32	0.00	No
Williamson Tea Kenya	Agriculture	1443	78.13	0.00	No
Olympia	Investment	1420	76.88	0.00	No
Umeme	Energy and Petroleum	1408	76.23	0.00	No
TPS Serena	Commercial and services	1355	73.36	0.00	No
Sanlam Kenya PLC	Insurance	1271	68.81	0.00	No
Crown Paints	Construction and Allied	1168	63.24	0.00	No
Standard Group Ltd	Commercial and services	1117	60.48	0.00	No
Unga Group	Manufacturing and Allied	1017	55.06	0.00	No
Kenya Airways	Commercial and services	967	52.36	0.00	No
Express Kenya	Commercial and services	919	49.76	0.00	No
East African Portland Cement	Construction and Allied	892	48.29	0.00	No
Eaagads	Agriculture	850	46.02	0.00	No
Car and General (Kenya)	Automobiles and Accessories	790	42.77	0.00	No
BOC Kenya	Manufacturing and Allied	760	41.15	0.00	No
Kakuzi	Agriculture	496	26.85	0.00	No
Kapchorua Tea Kenya	Agriculture	468	25.34	0.00	No
Limuru Tea	Agriculture	115	6.23	0.00	No
Africa Mega Agricornp	Manufacturing and Allied	49	2.65	0.00	No

Note: eligibility requires development coverage $\geq 90\%$ and missing-volume share $\leq 10\%$. The missing-volume criterion is nonbinding in the development-period eligibility audit; all exclusions are attributable to the coverage rule. Selection uses no information from the locked 2024-2026 test period.

A.3 Composition of the retained 29-firm sample

The retained sample spans nine sectors. Banking is the largest group with eight firms, followed by Commercial and services with six, Insurance with four, and Manufacturing and Allied and Energy and Petroleum with three each. The retained sample also includes Investment, Telecommunication and

Technology, Investment Services, and Agriculture. The analysis remains at the stock level throughout and does not use sector labels to construct synthetic OHLC series.

Table S3. Sector composition of the retained sample

Sector	Firms	Share of retained sample (%)
Banking	8	27.6
Commercial and services	6	20.7
Insurance	4	13.8
Manufacturing and Allied	3	10.3
Energy and Petroleum	3	10.3
Investment	2	6.9
Telecommunication and Technology	1	3.4
Investment Services	1	3.4
Agriculture	1	3.4

A.4 OHLC and volume data-quality rules

The preprocessing protocol distinguishes invalid or unavailable observations from economically meaningful non-trading. The study classifies an OHLC row as range-valid only when the reported high is at least as large as both the open and close and the reported low is no greater than either the open or close. The study excludes the 135 raw rows that violate this condition from range-based volatility construction and does not mechanically repair their prices.

$$\text{High}_t \geq \max(\text{Open}_t, \text{Close}_t) \quad \text{and} \quad \text{Low}_t \leq \min(\text{Open}_t, \text{Close}_t)$$

The study treats volume records represented by textual missing markers as missing values. By contrast, the study retains numeric zero-volume records as genuine observations of zero trading activity. This distinction matters in a thinly traded market because a zero can convey economically relevant liquidity information, whereas a missing marker represents data unavailability. Across the raw panel, 977 rows have missing Volume, and 876 have genuine numeric zero Volume.

A.5 Corporate-action segmentation

The study treats issued-share changes and recorded bonus-issue events as corporate-action boundaries for path-dependent features. At each identified event, a new feature segment begins. Rolling volatility variables and technical indicators restart after the event, and a future h-day Yang-Zhang target is unavailable if its forecast window crosses the event. The observed OHLC series is not mechanically back-adjusted using the issued-share change ratio because such changes need not map one-for-one into a price split adjustment.

The eligible sample contains 14 identified corporate-action events affecting 8 firms. Eleven events occur during the development period, and three occur during the locked test. Nairobi Business Ventures has the largest concentration, with four identified events.

Table S4. Corporate-action events in the eligible sample

Firm	Date	Period	Previous shares	Issued shares	Relative change	Recorded bonus issue
Flame Tree Group	2017-06-29	Development	161,866,804	178,053,486	0.1000	16,186,680
Co-operative Bank (Kenya)	2017-07-17	Development	4,889,312,246	5,867,174,695	0.2000	977862449
JUBILEE HOLDINGS	2017-08-03	Development	65,884,490	72,472,950	0.1000	6588460
Nairobi Business Ventures	2018-12-03	Development	23,600,000	38,600,000	0.6356	15,000,000
Kenya Re	2019-08-05	Development	699,949,068	2,799,796,272	3.0000	2,099,847,204
NCBA Bank	2019-10-01	Development	703,940,032	1,497,745,030	1.1277	793,804,998
NCBA Bank	2020-05-12	Development	1,497,745,030	1,647,519,532	0.1000	149,774,502

Firm	Date	Period	Previous shares	Issued shares	Relative change	Recorded bonus issue
Nairobi Business Ventures	2020-11-16	Development	38,600,000	77,200,000	1.0000	38,600,000
Nairobi Business Ventures	2020-11-30	Development	77,200,000	492,200,000	5.3756	415,000,000
Nairobi Business Ventures	2021-10-18	Development	492,200,000	1,353,711,934	1.7503	861,511,934
Nairobi Securities Exchange	2022-01-03	Development	259,500,791	260,004,320	0.0019	503529
Kenya Re	2024-08-26	Locked test	2,799,796,272	5,599,592,544	1.0000	2,799,796,272
CIC Insurance Group	2025-06-19	Locked test	2,615,578,472	926,955,357	0.6456	261,513,643
Nairobi Securities Exchange	2026-01-02	Locked test	260,004,320	260,897,735	0.0034	263194

Note: the event table audits segment-reset dates used in feature and target construction. The study calculates relative changes from issued shares and does not use these ratios to mechanically back-adjust OHLC prices.

A.6 multi-day target availability

Target availability decreases mechanically as the forecast horizon lengthens because a longer future window requires more valid OHLC observations and is more likely to intersect a corporate-action boundary or the end of the sample. The locked test contains 18,441 observed stock-date forecast origins before target-validity restrictions. Of these, 17,547 have a valid 5-day Yang-Zhang target, 16,885 have a valid 10-day target, and 15,787 have a valid 20-day target.

Table S5. Yang-Zhang target availability by forecast horizon

Horizon	Development origins with target	Locked origins observed	Locked origins with target	Locked target rate (%)
5 days	48,831	18,441	17,547	95.15
10 days	46,286	18,441	16,885	91.56
20 days	42,764	18,441	15,787	85.61

The corresponding locked target-validity rates are 95.15%, 91.56% and 85.61% for the 5-, 10- and 20-day horizons. No unavailable target is imputed. Model-specific common-support counts can be lower because a forecast is retained for a particular comparison only when all required model specifications produce finite predictions for the same stock-date origin.

A.7 Audit implications for the main analysis

The sample-construction and data-quality rules distinguish among listing and history coverage, economically meaningful non-trading, and genuine data unavailability. The 90% development-period coverage criterion removes securities that lack sufficient continuous history for the main rolling-feature analysis. The study retains genuine zero trading activity so that these observations continue to capture liquidity variation. The study also leaves invalid OHLC records and missing volume observations unrepaired rather than introducing mechanically imputed values.

Most importantly, the study does not select the 29-firm sample because those firms forecast well. The framework makes the eligibility decision before the locked test, using only data coverage and missingness conditions. This design prevents sample-selection look-ahead and preserves the locked 2024- 2026 period as an untouched evaluation sample.

A.8 Reproducibility note

The R pipeline generates Tables S1–S5 directly from the frozen preprocessing audit files. These files comprise `eligible_stock_audit.csv`, `preprocessing_audit_overall.csv`, `run_configuration.csv`, `preprocessing_audit_targets.csv`, and `corporate_action_audit.csv`. The replication package retains these audit files so that researchers can reconstruct the sample without inferring selection rules from the final forecasting panel.

Supporting Information B

Variable Definitions, Feature Construction and Technical-Indicator Formulas

B.1 Notation and construction conventions

For stock i and market date t , O , H , L and C denote open, high, low and close prices; V denotes reported trading volume; S denotes issued shares; and MC denotes market capitalisation. Log price ratios are defined only when both numerator and denominator are finite and strictly positive. The study computes path-dependent historical features within stock-specific post-corporate-action segments and excludes pre-event observations from post-event rolling variables.

The study calculates historical technical indicators and Yang–Zhang memory variables using observed stock sessions within each current corporate-action segment. In contrast, the study aligns future volatility targets with the full NSE market calendar. Hence, a 5-day target always refers to the next five NSE market dates rather than the next five observed records of an individual stock. The alignment therefore prevents an illiquid stock from extending a nominal 5-day target over a substantially longer calendar interval. Twenty-observation liquidity rolling means require at least 16 non-missing observations (80% of the window). The study does not impute missing targets in preprocessing. Model-specific imputation or scaling, where required, is estimated from the training sample only and is documented separately in Supporting Information C.

B.2 Yang-Zhang volatility components and targets

The target is the Yang-Zhang estimate of average daily variance over a future h -market-day window, with $h \in \{5, 10, 20\}$. The construction follows Yang and Zhang (2000) and combines overnight variance, open-to-close variance, and the Rogers-Satchell range component.

$$\begin{aligned} o_{i,t} &= \ln \left(\frac{O_{i,t}}{C_{i,t-1}} \right) \\ c_{i,t} &= \ln \left(\frac{C_{i,t}}{O_{i,t}} \right) \\ u_{i,t} &= \ln \left(\frac{H_{i,t}}{O_{i,t}} \right), \quad d_{i,t} = \ln \left(\frac{L_{i,t}}{O_{i,t}} \right) \\ RS_{i,t} &= u_{i,t}(u_{i,t} - c_{i,t}) + d_{i,t}(d_{i,t} - c_{i,t}) \end{aligned}$$

For a window of n observations, let s_o^2 and s_c^2 denote the usual sample variances of o and c (denominator $n - 1$), and let $\overline{RS}$ denote the arithmetic mean of the n Rogers-Satchell terms. The estimator used by the pipeline is

$$YZ_n = s_o^2 + k_n s_c^2 + (1 - k_n) \overline{RS},$$

where

$$k_n = \frac{0.34}{1.34 + \frac{n+1}{n-1}}.$$

The study floors tiny negative component estimates at zero when they arise solely from floating-point cancellation. The historical memory variables `yz_hist_5`, `yz_hist_10`, and `yz_hist_20` use the current and

preceding observed sessions within the same corporate-action segment. The future targets `target_yz_5`, `target_yz_10`, and `target_yz_20` use the next h NSE market dates on the full calendar.

The forward target is available only when all overnight, open-to-close, and Rogers-Satchell components are finite throughout the future window. A target is also invalidated when its window spans a corporate-action event. This strict rule explains why target availability decreases from 95.15% at 5 days to 85.61% at 20 days in the locked sample.

B.3 Frozen nested information sets

The forecasting experiment uses three nested information sets. Information Set A contains only multi-scale Yang-Zhang volatility memory. Information Set B adds nine underlying market variables. The primary Information Set B deliberately excludes `ret_oc`, although the study constructs that variable for the Yang-Zhang decomposition. The study also excludes `obs_share20` from the primary set and retains it only for liquidity and staleness robustness analysis.

Table S6. Frozen A-B-C feature manifest

Information set/family	Variable	Meaning	Main-model role
A: volatility memory	<code>yz_hist_5</code>	Trailing 5-session Yang-Zhang variance	A, B and C
A: volatility memory	<code>yz_hist_10</code>	Trailing 10-session Yang-Zhang variance	A, B and C
A: volatility memory	<code>yz_hist_20</code>	Trailing 20-session Yang-Zhang variance	A, B and C
B: underlying market variables	<code>ret_cc</code>	Close-to-close log return	B and C
B: underlying market variables	<code>abs_ret_cc</code>	Absolute close-to-close log return	B and C
B: underlying market variables	<code>ret_on</code>	Overnight log return	B and C
B: underlying market variables	<code>log_hl_range</code>	Log high-low price range	B and C
B: underlying market variables	<code>log_volume</code>	$\log(1 + \text{volume})$	B and C
B: underlying market variables	<code>log_turnover20</code>	$\log(1 + 20\text{-session mean turnover})$	B and C
B: underlying market variables	<code>zero_ret20</code>	20-session zero-return share	B and C
B: underlying market variables	<code>log_amihud20</code>	Logged/scaled 20-session Amihud measure	B and C
B: underlying market variables	<code>log_mcap</code>	Log market capitalisation	B and C
C: trend	<code>sma_gap20</code>	Price relative to 20-session SMA	C
C: trend	<code>nmacd_12_26</code>	Normalized EMA(12)-EMA(26) spread	C
C: trend	<code>adx14</code>	14-period average directional index	C
C: momentum	<code>rsi14</code>	14-period relative strength index	C
C: momentum	<code>roc10</code>	10-period rate of change	C
C: momentum	<code>stoch_k14</code>	14-period fast stochastic %K	C
C: range	<code>natr14</code>	14-period ATR normalised by close	C
C: range	<code>bbw20</code>	20-period Bollinger-band width	C

B.4 Underlying market variables

Underlying market variables are direct transformations or short rolling summaries of prices, trading activity, liquidity, and size. These variables provide a strong information benchmark before the study introduces technical indicators.

Table S7. Underlying market variable definitions

Variable	Exact construction	Window/validity	Interpretation
<code>ret_cc</code>	$\ln(C_t / C_{\text{prev}})$	Previous observed close within current segment	Close-to-close return
<code>abs_ret_cc</code>	$ \text{ret_cc} $	Same validity as <code>ret_cc</code>	Return magnitude

Variable	Exact construction	Window/validity	Interpretation
ret_on	$\ln(O_t / C_{prev})$	Previous observed close within current segment	Overnight return
log_hl_range	$\ln(H_t / L_t)$	Requires valid positive OHLC	Intraday range
log_volume	$\ln(1 + V_t)$	Missing V -> missing; numeric zero retained	Trading activity
turnover	V_t / S_t	Finite V and positive issued shares	Daily share turnover; supporting underlying market variable
turnover20	mean(turnover)	20 observed sessions; ≥ 16 non-missing	Rolling turnover
log_turnover20	$\ln(1 + \text{turnover20})$	turnover20 ≥ 0	Primary turnover input/liquidity measure
zero_ret20	mean[1(ret_cc=0)]	20 observed sessions; ≥ 16 non-missing	Staleness/non-trading proxy
amihud_day	$ ret_{cc} / (C_t V_t)$	Requires C>0 and V>0	Daily price-impact proxy
amihud20	mean(amihud_day)	20 observed sessions; ≥ 16 non-missing	Rolling Amihud measure
log_amihud20	$\ln(1 + 10^6 \times \text{amihud20})$	amihud20 ≥ 0	Numerically scaled logged Amihud input
log_mcap	$\ln(MC_t)$	Finite positive market capitalisation	Firm size
ret_oc	$\ln(C_t / O_t)$	Requires valid positive OHLC	Constructed for Yang-Zhang; not in primary B

B5 Technical-indicator definitions

The primary technical block contains eight price- and range-derived indicators. The study excludes volume-based technical trading rules from Information Set C. Information Set B already contains turnover and Amihud measures, so additional volume indicators could mechanically confound the liquidity analysis. All indicators restart at corporate-action segment boundaries. The ADX, RSI, and ATR constructions follow the technical-analysis conventions introduced by Wilder (1978), while Bollinger-band width follows the conventional two-standard-deviation bands described by Bollinger (2001).

Table S8. Technical-indicator construction

Family	Variable	Construction used	Lookback	Scale/note
Trend	sma_gap20	$C_t / \text{SMA}_{20}(t-1)$	20	SMA is a rolling mean of Close
Trend	nmacd_12_26	$[\text{EMA}_{12}(C)_t - \text{EMA}_{26}(C)_t] / C_t$	12, 26	Normalised MACD spread; no signal-line input
Trend	adx14	TTR: ADX(H,L,C,n=14), ADX output	14	Trend-strength index
Momentum	rsi14	TTR::RSI(C,n=14)	14	Relative strength index
Momentum	roc10	$C_t / C_{t-10} - 1$	10	Simple rate of change
Momentum	stoch_k14	TTR::stoch(H,L,C,nFastK=14, nFastD=3,nSlowD=3), fastK	14	Fast stochastic %K
Range	natr14	TTR::ATR(H,L,C,n=14)[ATR] / C_t	14	Average true range normalised by price
Range	bbw20	$4 \times \text{sd}_{20}(C) / \text{SMA}_{20}(C)$	20	Equivalent to (upper-lower)/middle for +/-2 sd bands

For clarity, the normalised MACD feature uses only the difference between the 12- and 26-period exponential moving averages divided by the contemporaneous Close; it does not include the common 9-period MACD signal line. The stochastic feature is the fast %K output. The study calculates Bollinger width using four rolling standard deviations because the conventional upper and lower bands lie two standard deviations above and below the 20-period moving average. The study calculates ADX, RSI, stochastic %K, and ATR using the TTR implementation and the parameters reported in Table S8. These settings fix the smoothing conventions used for replication.

B.6 Liquidity variables and ex-ante liquidity states

The study uses log_turnover20 as the primary liquidity-conditioning variable. At each NSE market date, the analysis ranks stocks with finite log_turnover20 values cross-sectionally. When at least 10 stocks are

available, the study uses average ranks to divide the cross-section into approximately equal terciles. It assigns stocks with ranks at or below $N/3$ to the Low state and stocks with ranks above $2N/3$ to the High state. All remaining stocks enter the Medium state. The classification uses only information available at the forecast origin.

The study retains two additional liquidity/staleness measures for robustness. `zero_ret20` is already part of B and captures the fraction of recent observed-session returns equal to zero. The study calculates `obs_share20` on the full NSE market calendar as the proportion of the preceding 20 market dates with an observed stock record. `log_amihud20` is also part of B and provides a price-impact proxy following the logic of Amihud (2002). These variables capture distinct dimensions of liquidity. Turnover measures trading intensity, while zero returns and observation share reflect trading continuity and potential staleness. The Amihud measure instead captures the magnitude of price movements relative to monetary trading volume.

Table S9. Liquidity and staleness variables

Variable	Definition	Calendar basis	Use
<code>log_turnover20</code>	$\ln(1 + \text{mean}(V/S))$ over 20 observed sessions	Observed sessions within segment	Primary B input and primary liquidity state
<code>zero_ret20</code>	Share of <code>ret_cc</code> equal to zero over 20 observed sessions	Observed sessions within segment	B input; liquidity robustness
<code>log_amihud20</code>	$\ln(1 + 10^6 \times \text{mean}[\text{ret_cc} /(C \cdot V)])$	Observed sessions within segment; $V > 0$	B input; liquidity robustness
<code>obs_share20</code>	Mean observed <code>row</code> over preceding 20 NSE market dates	Full market calendar	Staleness/coverage robustness only

B.7 Missingness, zeros and corporate-action resets

The construction rules deliberately distinguish an economic zero from missing data. The study retains numeric zero-volume observations, sets `log_volume` to zero, and assigns daily turnover a value of zero when issued-share data are valid. By contrast, the daily Amihud measure is unavailable when $V=0$ because its denominator would be zero. The study treats textual missing-volume markers as missing values and does not replace them during feature construction.

The study does not repair invalid OHLC rows. When $H < \max(O, C)$ or $L > \min(O, C)$, range-dependent variables including `ret_oc`, `log_hl_range`, and the Rogers-Satchell components are unavailable. If the previous Close within a segment is unavailable or non-positive, `ret_cc` and `ret_on` are unavailable. Technical indicators that require a complete lookback naturally remain missing until enough post-event history has accumulated.

The study applies corporate-action segmentation before constructing all path-dependent historical features. Thus SMA, EMA, ROC, ADX, RSI, stochastic, ATR, Bollinger width, and historical Yang-Zhang memory never use pre-event observations after a reset. Forward targets that cross an event are invalidated rather than mechanically adjusted.

B.8 Reproducibility audit

The frozen `feature_manifest.csv` confirms the model-facing variable families used in the completed run. The R preprocessing script constructs the full set once, and the exported `COLAB_FROZEN_PANEL.csv.gz` passes the same variables and targets to the Python LightGBM/LSTM stage without reconstructing them. This design ensures that differences across forecasting architectures reflect model behaviour rather than feature implementation differences. The exact primary B extras in the completed experiment are: `ret_cc`, `abs_ret_cc`, `ret_on`, `log_hl_range`, `log_volume`, `log_turnover20`, `zero_ret20`, `log_amihud20`, and `log_mcap`. The exact C additions are: `sma_gap20`, `nmacd_12_26`, `adx14`, `rsi14`, `roc10`, `stoch_k14`, `natr14` and `bbw20`. Treat these lists as frozen during manuscript revision.

Supporting Information C

Forecasting Protocol, Model Specifications, Hyperparameter Selection and Refit Design

C.1 Chronological development, validation and locked evaluation

All model development follows a chronological design. The common development sample begins on 1 August 2016. The study uses four expanding validation folds: training through 2019 and validating in 2020; training through 2020 and validating in 2021; training through 2021 and validating in 2022; and training through 2022 and validating in 2023. The final pseudo-out-of-sample interval from 2 January 2024 to 31 July 2026 remains locked against sample selection, feature selection, architecture choice, and hyperparameter tuning.

Because the dependent variable represents future variance over an h-day horizon, the study does not use a simple $\text{Date} < \text{refit date}$ rule to determine training eligibility. For each horizon, the study retains a training origin only when the final date of its target window falls strictly before the next refit or validation start. The purging rule prevents future target windows from overlapping fold boundaries.

Training origin eligible iff $\text{Date}_t < R$ and $\text{TargetEnd}_t(h) < R$.

Table S10. Expanding validation fold registry

Fold	Train start	Train end	Validation starts	Validation end
fold_2020	2016-08-01	2019-12-31	2020-01-01	2020-12-31
fold_2021	2016-08-01	2020-12-31	2021-01-01	2021-12-31
fold_2022	2016-08-01	2021-12-31	2022-01-01	2022-12-31
fold_2023	2016-08-01	2022-12-31	2023-01-01	2023-12-29

The calendar registry defines the nominal fold boundaries. The study further purges horizon-specific training samples by requiring $\text{TargetEnd}(h) < \text{validation/refit start}$.

C.2 Refit cadence by forecasting architecture

The study follows a recursive refitting procedure. After fixing the model definitions and tuning choices, the analysis incorporates newly observed outcomes at each scheduled refit date. Refit cadence differs by architecture because estimation cost and state evolution differ, but A, B, and C always share the same cadence within a given architecture. The central B-versus-C comparison is therefore not confounded by different update frequencies within the model.

Table S11. Forecast-model refit schedule

Architecture	Validation cadence	Locked-test cadence	Locked-test periods	Outcome-availability rule
Log-HAR-X	Annual validation folds	Monthly	31 locked-test monthly periods	Targets must fully end before refit
LightGBM	Annual validation folds	Annual	2024, 2025, 2026	B-tuned hyperparameters frozen for A/C/ablations
Pooled LSTM	Annual validation folds	Annual	2024, 2025, 2026	B-tuned architecture/epochs frozen
GJR-GARCH(1,1)-t	Quarterly from 2020	Quarterly	11 locked-test quarters/partial quarters	Return-state benchmark; no h-day target used in estimation

The R registry contains 35 Log-HAR periods (4 validation and 31 locked-test periods) and 27 GJR periods. The Python stage uses three annual locked-test periods: 2024, 2025 and January-July 2026.

C.3 Positivity-preserving Log-HAR-X

The econometric architecture is a pooled stock-fixed-effect regression estimated separately for each horizon, information set, and refit period. The response is the logarithm of scaled future Yang-Zhang variance:

$$y^*_{i,t+h} = \ln[\max(YZ_{i,t+1:t+h} \times 10,000, 0) + 10^{-8}].$$

The study transforms the three historical Yang–Zhang predictors to the same log-scaled variance scale. Return and range variables (`ret_cc`, `abs_ret_cc`, `ret_on`, and `log_hl_range`) are multiplied by 100; other underlying market and technical variables enter on their constructed scales. A stock factor supplies firm-specific intercepts.

$$y^*_{i,t+h} = \alpha_i + \beta_h' X_{i,t} + \epsilon_{i,t+h}.$$

The estimation uses complete training rows and an explicitly constructed model matrix. Before OLS estimation with `lm.fit()`, the study applies a QR decomposition with a tolerance of 10^{-10} to retain a full-rank subset of predictors. The QR screening prevents prediction problems caused by aliased coefficients. In the completed run, all 630 Log-HAR-X fits succeeded, and the procedure dropped no design columns for collinearity.

Level forecasts use a training-only Duan retransformation. The global smearing factor is the mean of $\exp(\epsilon)$ over training residuals after numerical clipping at ± 50 on the log scale. Use a stock-specific factor when that stock has at least 100 usable training residuals; otherwise, use the global factor.

Hence:

$$\text{Forecast}(YZ_i) = \exp(\text{predicted log variance}_i) \times \text{smear}_i / 10,000.$$

Positivity follows from exponentiation rather than from clipping negative forecasts. A floor of 10^{-12} exists only as an underflow guard. The positivity audit records no non-positive or floor-clipped forecasts in the completed run.

C.4 LightGBM tuning and locked forecasting

The study trains LightGBM on the same log-scaled target used in the nonlinear Python stage and supplies stock identity as a categorical feature with a globally fixed category set. The study searches candidate hyperparameters using Information Set B only. For each horizon, the analysis evaluates every candidate across the four expanding validation folds and selects the candidate with the lowest mean validation QLIKE. During tuning, the study allows boosting for up to 2,000 rounds and applies early stopping after 75 rounds without validation improvement. For the locked test, the study sets the number of boosting rounds to the median best iteration of the winning candidate across folds, subject to a minimum of 10 rounds.

The study requires complete contemporaneous predictors for every LightGBM origin row in the relevant information set and does not impute LightGBM inputs. The training procedure uses deterministic seeds of 42 for the model, feature fraction, bagging, and data sampling. It optimises regression loss on the log target and uses QLIKE for validation selection. The study then applies the hyperparameters selected under Information Set B unchanged to Information Sets A and C and to all indicator-family ablations.

Table S12. LightGBM candidate hyperparameter grid

Candidate	Learning rate	Leaves	Min leaf	Feature frac.	Bagging frac.	Bagging freq.	L2
LGB1	0.05	15	100	0.90	0.90	1	1.0
LGB2	0.05	31	100	0.90	0.90	1	1.0
LGB3	0.03	31	50	0.90	0.90	1	1.0

Candidate	Learning rate	Leaves	Min leaf	Feature frac.	Bagging frac.	Bagging freq.	L2
LGB4	0.03	63	100	0.80	0.90	1	2.0
LGB5	0.05	31	200	0.75	0.90	1	2.0
LGB6	0.03	15	200	1.00	1.00	0	1.0

Table S13. Selected LightGBM configuration by horizon

h	Winner	Mean validation QLIKE	Fixed rounds	LR	Leaves	Min leaf	Feature frac.	L2
5	LGB5	0.413084	36	0.05	31	200	0.75	2.0
10	LGB5	0.279424	38	0.05	31	200	0.75	2.0
20	LGB5	0.197852	40	0.05	31	200	0.75	2.0

The study selects LGB5 at all three horizons and fixes the locked-test round counts at 36, 38, and 40 for the 5-, 10-, and 20-day forecasts, respectively.

LightGBM forecasts are exponentiated and retransformed with the same training-only global/stock-specific Duan smearing rule used in the Python pipeline. A stock-specific smearing factor requires at least 100 training residuals; otherwise the global factor is used. Locked-test models are re-estimated annually with expanding data whose h-day outcomes have fully ended before the refit date.

C.5 Pooled LSTM sequence construction, scaling and tuning

The LSTM uses a 20-observation sequence ending at the forecast origin. Sequence indices are constructed separately within each Stock x corporate-action segment, so no sequence crosses a segment reset. A sequence is available only after 20 observations have accumulated within the current segment. At the forecast origin itself, every feature in the information set must be finite.

The procedure permits missing values only at earlier steps within an otherwise eligible sequence. The study imputes those values using feature medians estimated solely from the training sequences. It then standardises each feature using the corresponding training-only mean and standard deviation. The analysis applies the same medians, means, and standard deviations to the validation and test sequences. Using training-only imputation and scaling parameters preserves chronology while retaining sequences with occasional earlier missing observations.

Each candidate network contains one LSTM layer and a learned stock embedding. The architecture concatenates the LSTM state with the embedding, passes the combined representation through a ReLU dense layer and dropout, and produces a single linear log-variance output. The study trains each model with the Adam optimiser and MSE loss on the log-scaled target. The model-selection procedure monitors validation QLIKE. Development training allows up to 50 epochs, applies early stopping after six epochs without improvement, and restores weights from the best validation epoch.

The study shuffles training batches only within the pre-defined training sample. Batch shuffling does not alter the chronological fold structure because all validation and test dates remain fixed. The study sets all random seeds to 42 and requests deterministic TensorFlow operations where available.

Table S14. LSTM candidate architecture grid

Candidate	LSTM units	Dense units	Dropout	Learning rate	Batch size	Embedding dim.
LSTM1	32	16	0.10	0.0010	512	4
LSTM2	64	32	0.20	0.0007	512	4
LSTM3	48	24	0.10	0.0005	512	6

Table S15. Selected LSTM configuration by horizon

h	Winner	Mean validation QLIKE	Fixed epochs	LSTM units	Dense units	Dropout	Learning rate	Embed.
5	LSTM1	0.434558	2	32	16	0.10	0.0010	4
10	LSTM1	0.301662	2	32	16	0.10	0.0010	4

h	Winner	Mean validation QLIKE	Fixed epochs	LSTM units	Dense units	Dropout	Learning rate	Embed.
20	LSTM3	0.227418	2	48	24	0.10	0.0005	6

The study selects LSTM1 at the 5- and 10-day horizons and LSTM3 at the 20-day horizon. Because the median best development epoch falls below two for some selections, the pre-specified lower bound sets locked-test training to two epochs.

Across the 29 retained firms, the sequence-availability rate has a median of 99.23% and ranges from 96.54% to 99.24%. Supporting Information Section G reports the full stock-level sequence-availability diagnostics alongside the model-diagnostic tables. After fixed-epoch estimation, the study retransforms LSTM log predictions using training-only global and stock-specific Duan smearing. The procedure requires at least 100 observations to estimate a stock-specific factor. During the locked test, the study applies the architecture selected under Information Set B unchanged to Information Sets A and C and to all indicator-family ablations.

C.6 GJR-GARCH(1,1)-Student-t conventional benchmark

The study estimates the conventional benchmark separately for each stock using close-to-close log returns multiplied by 100. The conditional mean specification includes an intercept and no ARMA terms. The study models conditional variance with a GJR-GARCH(1,1) specification with standardised Student-t innovations and estimates the model in rugarch using the hybrid solver. At each refit date, the estimation requires at least 500 finite training returns.

$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \gamma I(\epsilon_{t-1} < 0) \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2$. After estimation at a quarterly boundary, the variance state is updated recursively through the quarter using returns observed by each forecast origin. The persistence used for multi-step expected variance is $\alpha + \beta + \gamma/2$. The h-day benchmark forecast is the mean of the next h expected daily conditional variances, converted from percentage-return variance back to raw log-return variance. The conversion places the benchmark forecast on the same scale as the Yang–Zhang average daily variance target. All 783 GJR fit attempts in the completed registry returned status 'ok'. Supporting Information G reports detailed stock-level benchmark behaviour, including thin-trading pathologies that affect pooled validation QLIKE.

C.7 Common target transformation and retransformation safeguards

The Python models use $\log(\max(YZ \times 10,000, 0) + 10^{-8})$ as the training target, matching the scale convention used by Log-HAR-X. Exponentiation guarantees a positive level before division by 10,000. Training-only Duan smearing corrects the Jensen retransformation bias. For all three A/B/C specifications, the study estimates the smearing factor separately at each model refit and excludes future residuals from the correction. The central feature-comparison safeguard is that tuning occurs on B only for the nonlinear architectures. Technical-indicator specification C therefore receives no additional hyperparameter search. Keeping the tuning procedure fixed prevents any apparent B-to-C improvement from arising simply because the larger feature set receives more tuning opportunities.

C.8 Software environment, reproducibility and runtime

The study conducts the preprocessing, Log-HAR-X, and GJR stages in R. It runs the LightGBM and LSTM stages in Python/Google Colab using the frozen panel exported from R. The Python stage does not reconstruct the targets or technical indicators. The study uses seed 42 throughout the Python analysis and the block-bootstrap procedures.

Table S16. Core software environment captured by the completed runs

Software/package	Version	Environment	Role
R	4.5.1	Windows 11 x64 (build 26200)	Core R pipeline
data.table	1.18.0	R	Panel construction / fast aggregation
TTR	0.24.4	R	Technical indicators
rugarch	1.5-4	R	GJR-GARCH estimation
zip	2.3.3	R	Output packaging
Python	3.12.13	Linux 6.6.122+ / glibc 2.35	ML/DL pipeline
NumPy	2.0.2	Python	Numerical arrays
pandas	2.2.2	Python	Panel/data handling
LightGBM	4.6.0	Python	Tree-based forecasting
TensorFlow	2.20.0	Python	Pooled LSTM

The R pipeline completed in 11.858 minutes. The Python ML/DL stage completed in 46.01 minutes. The Colab runtime reported GPU available = False; the completed run therefore executed on CPU. The study retains the full session and version manifests with the replication outputs.

C.9 Reproducibility checklist and locked-design safeguards

- The study does not use random train/test splitting; all validation and locked-test partitions are chronological.
- Training h-day targets must end before each validation/refit boundary.
- The study determines firm eligibility and feature definitions before opening the locked test sample.
- The study tunes LightGBM and LSTM hyperparameters using Information Set B only and applies the selected settings unchanged to Information Sets A and C and to all family ablations.
- LSTM imputation and scaling parameters are estimated from training sequences only.
- The study estimates log-to-level smearing factors using training residuals only.
- LSTM sequences never cross corporate-action segments.
- The Python stage consumes the frozen R feature/target panel rather than independently reconstructing variables.

Supporting Information D

Extended Forecast Results and Robustness Tables

D.1 Development-period evidence and attenuation in the locked sample

Log-HAR-X provides the clearest direct comparison between development validation and the final locked sample because it retained B and C forecasts in both stages. During the 2020-2023 validation period, the technical block lowers QLIKE by 5.54% at five days and 6.84% at 10 days, whereas the 20-day result is negative. The shorter-horizon gains shrink below one per cent in the locked 2024-2026 sample and lose statistical support. This attenuation is one reason the manuscript bases its conclusions on the locked sample rather than on development-period performance.

Table S17. Log-HAR-X B-versus-C performance: development validation versus locked test

Stage	h	N	QLIKE B	QLIKE C	TI gain (%)	Holm-HAC p	Holm-bootstrap p
Locked test 2024-2026	5	16,408	0.4267	0.4244	0.56	0.3377	0.3320
Locked test 2024-2026	10	15,780	0.2973	0.2946	0.93	0.3377	0.3320
Locked test 2024-2026	20	14,761	0.2008	0.2037	-1.47	0.0325	0.0294
Validation 2020-2023	5	25,427	0.4585	0.4331	5.54	0.0339	0.0564
Validation 2020-2023	10	24,200	0.3290	0.3065	6.84	0.0204	0.0300
Validation 2020-2023	20	22,527	0.2418	0.2469	-2.14	0.1622	0.1592

Positive *TI* gain means *C* lowers *QLIKE* relative to *B*. The validation five-day result is significant under Holm-HAC but not under Holm-bootstrap at 5%; the 10-day validation gain survives both procedures. In the locked sample, no positive gain survives either adjusted test, while the 20-day deterioration does.

D.2 Fold-level development performance of the selected nonlinear B models

For LightGBM and the pooled LSTM, the study selects nonlinear hyperparameters using Information Set B only. Table S18 reports the fold-by-fold development performance of the candidate selected for each horizon. The reported values document the tuning evidence available before the study opened the locked sample. The study does not conduct separate hyperparameter searches for Information Sets A or C.

Table S18A. Selected LightGBM B-model performance across development folds

h	Validation year	Selected candidate	QLIKE	MSE	Best round	Train N	Validation N
5	2020	LGB5	0.3990	0.0000051	29	20,734	6,320
5	2021	LGB5	0.4552	0.0000025	37	27,041	6,702
5	2022	LGB5	0.4046	0.0000024	35	33,758	6,240
5	2023	LGB5	0.3935	0.0000029	45	39,981	6,189
10	2020	LGB5	0.2634	0.0000030	36	19,450	6,037
10	2021	LGB5	0.3260	0.0000018	63	25,455	6,476
10	2022	LGB5	0.2687	0.0000014	34	31,966	5,857
10	2023	LGB5	0.2596	0.0000018	39	37,798	5,850
20	2020	LGB5	0.1939	0.0000021	38	17,622	5,654
20	2021	LGB5	0.2392	0.0000013	74	23,214	6,142
20	2022	LGB5	0.1741	0.0000008	36	29,409	5,360
20	2023	LGB5	0.1842	0.0000010	41	34,785	5,391

The selected LightGBM candidate is LGB5 at all three horizons.

Table S18B. Selected LSTM B-model performance across development folds

h	Validation year	Selected candidate	QLIKE	MSE	Best epoch	Train N	Validation N
5	2020	LSTM1	0.4001	0.0000057	3	20,734	6,320
5	2021	LSTM1	0.5158	0.0000041	1	27,041	6,702
5	2022	LSTM1	0.4005	0.0000027	2	33,758	6,240
5	2023	LSTM1	0.4218	0.0000035	1	39,981	6,189
10	2020	LSTM1	0.2730	0.0000037	2	19,450	6,037
10	2021	LSTM1	0.3862	0.0000033	1	25,455	6,476
10	2022	LSTM1	0.2640	0.0000016	2	31,966	5,857
10	2023	LSTM1	0.2835	0.0000023	1	37,798	5,850
20	2020	LSTM3	0.2525	0.0000113	2	17,622	5,654
20	2021	LSTM3	0.2516	0.0000016	4	23,214	6,142
20	2022	LSTM3	0.2021	0.0000017	2	29,409	5,360
20	2023	LSTM3	0.2035	0.0000015	2	34,785	5,391

The study selects LSTM1 at the 5- and 10-day horizons and LSTM3 at the 20-day horizon.

D.3 Locked-test pooled QLIKE and MSE on strict B-C common support

The main manuscript emphasises QLIKE because it is the primary volatility-forecast loss. Table S19 adds MSE on the identical B-C stock-date support. The secondary loss does not reveal a stable technical-indicator advantage. Log-HAR-X MSE worsens at all horizons, LightGBM is essentially unchanged, and LSTM improves only slightly at five days before worsening at 10 and 20 days.

Table S19. Locked-test pooled B-versus-C performance under QLIKE and MSE

Model	h	N	QLIKE B	QLIKE C	QLIKE gain (%)	MSE B	MSE C	MSE gain (%)
Log-HAR-X	5	16,408	0.4267	0.4244	0.56	0.000018	0.000019	-6.12
Log-HAR-X	10	15,780	0.2973	0.2946	0.93	0.000006	0.000006	-1.73

Model	h	N	QLIKE B	QLIKE C	QLIKE gain (%)	MSE B	MSE C	MSE gain (%)
Log-HAR-X	20	14,761	0.2008	0.2037	-1.47	0.000002	0.000003	-4.89
LightGBM	5	16,408	0.4191	0.4175	0.40	0.000010	0.000010	-0.31
LightGBM	10	15,780	0.2926	0.2897	0.96	0.000005	0.000005	0.07
LightGBM	20	14,761	0.1986	0.1971	0.77	0.000002	0.000002	-0.56
LSTM	5	16,408	0.4222	0.4293	-1.68	0.000010	0.000010	0.66
LSTM	10	15,780	0.2948	0.3017	-2.35	0.000006	0.000006	-0.17
LSTM	20	14,761	0.2067	0.2165	-4.75	0.000003	0.000003	-2.72

Positive gain means lower loss under C. All columns use exact B-C common support.

D.4 Equal-stock-weight robustness

Pooled loss weights firms with more forecast origins more heavily. An equal-stock-weight robustness check first computes each firm's average loss and then averages those stock-level losses. The qualitative message is unchanged: LightGBM retains small positive QLIKE gains, Log-HAR-X becomes negative at 20 days, and the LSTM deteriorates at every horizon.

Table S20. Equal-stock-weight locked-test B-versus-C robustness

Model	h	Stocks	Mean stock QLIKE B	Mean stock QLIKE C	QLIKE gain (%)	Mean stock MSE B	Mean stock MSE C	MSE gain (%)
Log-HAR-X	5	29	0.4332	0.4309	0.53	0.000021	0.000022	-6.17
Log-HAR-X	10	29	0.3041	0.3018	0.75	0.000007	0.000007	-1.35
Log-HAR-X	20	29	0.2114	0.2148	-1.59	0.000003	0.000003	-3.68
LightGBM	5	29	0.4262	0.4238	0.55	0.000011	0.000011	-0.24
LightGBM	10	29	0.2998	0.2969	0.98	0.000007	0.000007	0.11
LightGBM	20	29	0.2094	0.2065	1.38	0.000003	0.000003	-0.08
LSTM	5	29	0.4302	0.4378	-1.77	0.000012	0.000012	0.40
LSTM	10	29	0.3040	0.3117	-2.52	0.000007	0.000007	-0.50
LSTM	20	29	0.2197	0.2334	-6.27	0.000003	0.000003	-3.60

D.5 Cross-sectional breadth of technical-indicator gains

Aggregate QLIKE gains are not necessarily pervasive across firms. Table S21 therefore reports the number of stocks for which C improves on B, together with the mean and median of firm-specific percentage gains. LightGBM's pooled gain is positive at all horizons, yet fewer than half of the 29 firms improve at each horizon. The positive pooled result arises because a minority of relatively large firm-level improvements outweigh many smaller deteriorations in the aggregate loss.

Table S21. Stock-level breadth of B-to-C QLIKE gains

Model	h	Stocks	Improving	Deteriorating	Mean firm gain (%)	Median firm gain (%)
Log-HAR-X	5	29	13	16	0.13	-0.11
Log-HAR-X	10	29	16	13	0.20	0.56
Log-HAR-X	20	29	9	20	-3.05	-2.42
LightGBM	5	29	8	21	-0.41	-0.54
LightGBM	10	29	14	15	-0.35	-0.05
LightGBM	20	29	10	19	-0.68	-1.29
LSTM	5	29	10	19	-1.68	-1.80
LSTM	10	29	9	20	-2.18	-2.61
LSTM	20	29	9	20	-4.04	-4.66

Firm gain = $100 \times [\text{firm QLIKE}(B) - \text{firm QLIKE}(C)] / \text{firm QLIKE}(B)$, calculated on stock-specific common B-C support. Mean of firm percentage gains need not equal the percentage gain from the equal-stock-weight mean losses.

D.6 Complete stock-level B-to-C QLIKE gains

Tables S22A-S22C report the complete cross-section underlying the breadth counts. The same stock-date support count applies to all three architectures within a given stock and horizon because the frozen panel and B-C availability rules are shared.

Table S22A. Complete firm-level QLIKE gains at the 5-day horizon

Firm	N	Log-HAR-X gain (%)	LightGBM gain (%)	LSTM gain (%)
Absa Bank Kenya	636	-0.04	-0.96	-4.17
Britam Holdings	611	-1.24	-0.52	2.22
CIC Insurance Group	487	0.55	-0.21	1.26
Carbacid Investments Ltd	530	3.03	-3.42	-4.06
Centum	586	-0.99	-0.14	-0.46
Co-operative Bank (Kenya)	561	1.70	-0.23	-8.78
Diamond Trust Bank	636	5.51	0.18	0.17
East African Breweries	580	-0.90	-1.01	-6.16
Equity Group	611	2.77	-0.03	-2.77
Eveready	593	-0.15	-5.72	-9.45
Flame Tree Group	462	-2.16	-0.48	-7.83
HF Group	586	1.04	-5.47	-7.07
Home Africa	565	-1.69	-5.21	-2.67
JUBILEE HOLDINGS	436	1.79	4.02	-1.13
KCB Bank	582	4.20	0.51	-1.39
KPLC	611	-0.80	-1.35	-8.05
KenGen	610	1.25	-1.08	-4.01
Kenya Re	604	-2.23	-3.58	8.29
Longhorn Publishers	422	-3.35	-1.28	-2.28
NCBA Bank	611	3.78	-0.96	3.43
Nairobi Business Ventures	548	-0.11	4.39	5.61
Nairobi Securities Exchange	548	-1.66	-0.54	0.88
Nation Media Group	549	-0.99	-4.83	-8.20
Safaricom PLC	586	1.14	3.32	-0.84
Sasini	478	2.11	4.24	-3.71
Standard Chartered Bank	605	0.69	2.28	-1.80
Total Energy	540	-2.37	11.67	2.74
Uchumi Supermarkets	636	-6.79	-0.72	5.63
WWP Scangroup	598	-0.39	-4.79	5.78

Positive values indicate lower firm-level QLIKE after adding the full technical-indicator block.

Table S22B. Complete firm-level QLIKE gains at the 10-day horizon

Firm	N	Log-HAR-X gain (%)	LightGBM gain (%)	LSTM gain (%)
Absa Bank Kenya	631	0.90	0.84	-4.97
Britam Holdings	601	-0.48	2.01	2.07
CIC Insurance Group	462	2.35	1.51	-2.61
Carbacid Investments Ltd	505	5.51	-0.44	-4.73
Centum	571	-1.21	1.62	-3.03
Co-operative Bank (Kenya)	544	2.70	0.15	-10.30
Diamond Trust Bank	631	8.10	2.61	1.27
East African Breweries	560	-2.84	-4.74	-6.05
Equity Group	601	4.68	1.66	-1.81
Eveready	568	0.56	-4.84	1.76
Flame Tree Group	392	-0.09	0.39	-15.63
HF Group	571	1.99	-8.66	-12.37
Home Africa	545	-1.01	-1.74	-6.51
JUBILEE HOLDINGS	406	0.76	-6.81	6.10
KCB Bank	567	1.20	-4.26	-5.24
KPLC	601	2.56	-4.01	-12.66
KenGen	600	1.11	-0.66	-5.51
Kenya Re	594	2.38	-0.80	10.11

Firm	N	Log-HAR-X gain (%)	LightGBM gain (%)	LSTM gain (%)
Longhorn Publishers	387	-8.36	-4.95	-4.48
NCBA Bank	601	7.15	-2.59	3.21
Nairobi Business Ventures	538	-1.49	1.30	-1.60
Nairobi Securities Exchange	524	-3.76	-4.10	2.13
Nation Media Group	509	-0.22	-5.44	-5.79
Safaricom PLC	571	1.61	-0.05	-10.11
Sasini	427	-0.35	1.01	-1.59
Standard Chartered Bank	590	0.19	7.86	-1.73
Total Energy	485	-3.58	16.61	-1.18
Uchumi Supermarkets	631	-11.83	2.23	12.29
WWP Scangroup	567	-2.72	4.09	15.84

Positive values indicate lower firm-level QLIKE after adding the full technical-indicator block.

Table S22C. Complete firm-level QLIKE gains at the 20-day horizon

Firm	N	Log-HAR-X gain (%)	LightGBM gain (%)	LSTM gain (%)
Absa Bank Kenya	621	1.29	1.17	-7.74
Britam Holdings	581	1.92	-1.75	-0.36
CIC Insurance Group	412	-2.45	0.57	15.31
Carbacid Investments Ltd	455	2.37	-4.10	-4.66
Centum	541	-5.88	5.62	-6.44
Co-operative Bank (Kenya)	514	-2.67	-0.39	-2.08
Diamond Trust Bank	621	-0.15	-0.22	-8.85
East African Breweries	524	-10.65	-4.73	-4.72
Equity Group	581	5.10	9.00	-4.92
Eveready	525	-4.71	-9.96	-13.46
Flame Tree Group	305	-0.22	3.18	-82.62
HF Group	541	-1.31	-8.33	-10.47
Home Africa	505	-4.27	-8.11	-11.93
JUBILEE HOLDINGS	346	-3.93	5.39	10.97
KCB Bank	544	-0.26	-6.57	4.74
KPLC	581	4.36	-9.04	1.13
KenGen	580	1.76	-3.43	3.55
Kenya Re	574	0.80	-2.49	-1.39
Longhorn Publishers	364	-18.10	-1.29	-6.38
NCBA Bank	581	3.47	-0.83	-3.21
Nairobi Business Ventures	518	-5.56	9.66	13.75
Nairobi Securities Exchange	484	-2.46	-2.15	-12.23
Nation Media Group	461	5.83	5.71	-9.34
Safaricom PLC	541	-2.55	-6.91	-6.19
Sasini	352	-2.42	-1.39	8.90
Standard Chartered Bank	560	-3.08	2.21	-0.49
Total Energy	409	-2.01	15.05	-8.14
Uchumi Supermarkets	621	-32.88	-1.16	25.34
WWP Scangroup	519	-9.76	-4.43	4.89

Positive values indicate lower firm-level QLIKE after adding the full technical-indicator block.

D.7 Robustness interpretation

The extended evidence strengthens rather than changes the main manuscript interpretation. First, the development-to-locked comparison shows that apparently meaningful short-horizon Log-HAR-X gains attenuate substantially in the untouched future sample. Second, MSE provides no stable alternative-loss advantage for the technical block. Third, equal-stock weighting does not reverse the architecture-level pattern. Fourth, the complete stock cross-section shows that aggregate gains are sparse rather than

pervasive. Accordingly, the central conclusion does not depend on a single pooling convention, a single loss function, or a small subset of reported securities.

Supporting Information E

Technical-Indicator Family Ablations

E.1 Purpose and interpretation of the family ablations

The main manuscript tests the full technical-indicator augmentation in Information Set C against the underlying market-information benchmark in Information Set B. The family ablations ask a secondary question: whether any broad technical family accounts for the small pooled gains or deteriorations observed in the full B-versus-C experiment. Each ablation adds only one pre-specified family to B while holding the forecasting architecture and all previously selected model settings fixed.

The trend family contains the 20-session SMA gap, normalised MACD (12,26), and ADX (14). The momentum family includes RSI (14), ROC (10), and stochastic fast %K (14). The range and volatility family includes normalised ATR (14) and 20-session Bollinger-band width. Supporting Information Section B reports the exact formulas.

For LightGBM and the pooled LSTM, family-ablation models inherit the hyperparameters selected using B only; no family receives a separate tuning search. Log-HAR-X uses the same refit schedule and regression specification as the main experiment. The study therefore interprets family-specific QLIKE gains as descriptive changes in information content rather than as effects of additional optimisation.

E.2 Locked-test QLIKE results by technical-indicator family

Table S23 reports the locked-test QLIKE comparison for each family. Positive values indicate that adding the family lowers QLIKE relative to B on that family's exact common support. The results show a mixed pattern. Momentum provides the most frequent gains at the five- and 10-day horizons but deteriorates at 20 days across all three architectures. Range indicators are generally unfavourable at short horizons, with the main exception of the 20-day LightGBM result. Trend indicators produce only small positive gains in a subset of cells and deteriorate at the 20-day horizon under Log-HAR-X and LSTM.

Table S23. Locked-test QLIKE gains from individual technical-indicator families

Model	h	Family	N	QLIKE B	QLIKE B+family	QLIKE gain (%)
Log-HAR-X	5	Trend	16,408	0.4267	0.4264	0.08
Log-HAR-X	5	Momentum	16,422	0.4267	0.4236	0.74
Log-HAR-X	5	Range	16,422	0.4267	0.4270	-0.07
Log-HAR-X	10	Trend	15,780	0.2973	0.2973	0.03
Log-HAR-X	10	Momentum	15,794	0.2975	0.2950	0.84
Log-HAR-X	10	Range	15,794	0.2975	0.2977	-0.07
Log-HAR-X	20	Trend	14,761	0.2008	0.2027	-0.97
Log-HAR-X	20	Momentum	14,775	0.2010	0.2033	-1.16
Log-HAR-X	20	Range	14,775	0.2010	0.2025	-0.78
LightGBM	5	Trend	16,408	0.4191	0.4169	0.53
LightGBM	5	Momentum	16,422	0.4191	0.4160	0.73
LightGBM	5	Range	16,422	0.4191	0.4237	-1.10
LightGBM	10	Trend	15,780	0.2926	0.2932	-0.22
LightGBM	10	Momentum	15,794	0.2927	0.2911	0.56
LightGBM	10	Range	15,794	0.2927	0.2929	-0.06
LightGBM	20	Trend	14,761	0.1986	0.1982	0.18
LightGBM	20	Momentum	14,775	0.1988	0.2031	-2.16
LightGBM	20	Range	14,775	0.1988	0.1969	0.95

Model	h	Family	N	QLIKE B	QLIKE B+family	QLIKE gain (%)
LSTM	5	Trend	16,408	0.4222	0.4206	0.37
LSTM	5	Momentum	16,422	0.4222	0.4195	0.65
LSTM	5	Range	16,422	0.4222	0.4268	-1.10
LSTM	10	Trend	15,780	0.2948	0.3001	-1.78
LSTM	10	Momentum	15,794	0.2952	0.2949	0.10
LSTM	10	Range	15,794	0.2952	0.2984	-1.11
LSTM	20	Trend	14,761	0.2067	0.2137	-3.36
LSTM	20	Momentum	14,775	0.2073	0.2098	-1.21
LSTM	20	Range	14,775	0.2073	0.2123	-2.38

$Gain = 100 \times [QLIKE(B) - QLIKE(B+family)] / QLIKE(B)$. Positive values favour the family augmentation. The study matches support within each B-versus-family comparison; N may differ slightly across families because indicator availability differs.

E.3 Secondary MSE evidence

The secondary MSE results are less coherent than QLIKE and reinforce the decision not to promote any family as a robust source of incremental information. In particular, some short-horizon Log-HAR-X family additions produce large percentage MSE deteriorations despite modest QLIKE changes. Because volatility loss functions weight forecast errors differently, we report these discrepancies for transparency rather than using them to re-rank the families.

Table S24. Locked-test MSE gain from individual technical-indicator families

Model	h	Family	MSE gain (%)
Log-HAR-X	5	Trend	-5.95
Log-HAR-X	5	Momentum	-28.74
Log-HAR-X	5	Range	11.24
Log-HAR-X	10	Trend	0.12
Log-HAR-X	10	Momentum	0.09
Log-HAR-X	10	Range	-1.37
Log-HAR-X	20	Trend	0.21
Log-HAR-X	20	Momentum	-0.32
Log-HAR-X	20	Range	-4.00
LightGBM	5	Trend	0.10
LightGBM	5	Momentum	0.20
LightGBM	5	Range	-0.73
LightGBM	10	Trend	0.23
LightGBM	10	Momentum	0.16
LightGBM	10	Range	-0.05
LightGBM	20	Trend	-0.09
LightGBM	20	Momentum	-0.61
LightGBM	20	Range	-0.33
LSTM	5	Trend	0.65
LSTM	5	Momentum	1.31
LSTM	5	Range	0.67
LSTM	10	Trend	-0.63
LSTM	10	Momentum	1.09
LSTM	10	Range	0.19
LSTM	20	Trend	-3.55
LSTM	20	Momentum	-0.98
LSTM	20	Range	-2.12

Positive MSE gain means lower MSE after adding the indicated family. For Log-HAR-X, the values are recomputed directly from the frozen family forecasts on exact B-versus-family common support. QLIKE remains the primary loss.

E.4 Log-HAR-X family results in development validation versus the locked test

The study retained the Log-HAR-X family ablations for both development validation and the locked test. They show the same attenuation pattern as the full technical block. Trend gains of 2.76% and 3.26% at five and 10 days during validation shrink to 0.08% and 0.03% in the locked test. Momentum gains of

4.63% and 4.48% shrink to 0.74% and 0.84%. The validation range-family gain of 2.00% at 20 days reverses to a 0.78% deterioration.

Table S25. Log-HAR-X family-ablation attenuation from development validation to the locked test

Stage	h	Family	QLIKE B	QLIKE B+family	QLIKE gain (%)
Validation 2020-2023	5	Trend	0.4585	0.4458	2.76
Validation 2020-2023	5	Momentum	0.4592	0.4380	4.63
Validation 2020-2023	5	Range	0.4592	0.4559	0.72
Validation 2020-2023	10	Trend	0.3290	0.3183	3.26
Validation 2020-2023	10	Momentum	0.3296	0.3148	4.48
Validation 2020-2023	10	Range	0.3296	0.3271	0.75
Validation 2020-2023	20	Trend	0.2418	0.2423	-0.24
Validation 2020-2023	20	Momentum	0.2435	0.2572	-5.61
Validation 2020-2023	20	Range	0.2435	0.2386	2.00
Locked test 2024-2026	5	Trend	0.4267	0.4264	0.08
Locked test 2024-2026	5	Momentum	0.4267	0.4236	0.74
Locked test 2024-2026	5	Range	0.4267	0.4270	-0.07
Locked test 2024-2026	10	Trend	0.2973	0.2973	0.03
Locked test 2024-2026	10	Momentum	0.2975	0.2950	0.84
Locked test 2024-2026	10	Range	0.2975	0.2977	-0.07
Locked test 2024-2026	20	Trend	0.2008	0.2027	-0.97
Locked test 2024-2026	20	Momentum	0.2010	0.2033	-1.16
Locked test 2024-2026	20	Range	0.2010	0.2025	-0.78

E.5 Cross-architecture breadth of family-specific gains

Table S26 summarises how often each family produces a positive QLIKE gain across the nine locked model-horizon cells. Momentum is positive in six of nine cells, trend in five of nine, and range in only one of nine. However, none of the families is positive in all three architectures at the 20-day horizon, and only momentum and trend are positive in all three architectures at five days. This breadth pattern supports describing momentum as the most frequently favourable short-horizon family rather than as a robust technical-indicator explanation.

Table S26. Descriptive breadth of positive QLIKE family gains across architectures and horizons

Family	Positive cells	Negative cells	Mean gain (%)	Median gain (%)	Positive at 5d (out of 3)	Positive at 10d (out of 3)	Positive at 20d (out of 3)
Trend	5	4	-0.57	0.03	3	1	1
Momentum	6	3	-0.10	0.56	3	3	0
Range	1	8	-0.64	-0.78	0	0	1

This table is descriptive. The family-ablation analysis was not assigned a separate multiplicity-adjusted confirmatory testing family, so sign counts should not be interpreted as formal evidence of family-level significance.

E.6 Interpretation and role in the manuscript

The family evidence does not identify a stable technical source that reverses the main redundancy conclusion. Momentum indicators show the most consistent short-horizon numerical gains, but those gains weaken or reverse at 20 days and are not uniform across loss functions. Trend indicators show small, architecture-dependent effects. Range and volatility indicators are usually unfavourable despite their close relation to the volatility target, suggesting that the underlying range and volatility-memory variables already included in Information Set B absorb much of their information.

Accordingly, the main manuscript should use the family ablations only to explain the composition of the full C result. They should not be presented as new hypotheses or as evidence that a particular indicator family is statistically superior. This placement follows the broader forecasting practice of using feature-

group and ablation results as interpretive robustness evidence while keeping the primary forecast comparison focused.

Supporting Information F

Liquidity-Conditioned Technical-Indicator Value and Formal State Contrasts

F.1 Primary liquidity-state construction

The primary state variable is the 20-session rolling turnover measure, $\log_turnover_{20}$. At each NSE market date, stocks with a finite turnover measure are ranked cross-sectionally and divided into approximately equal Low, Medium, and High liquidity terciles. Because the classification is date-specific, the states represent relative liquidity within the contemporaneous cross-section rather than fixed firm types. State assignment uses information available at the forecast origin only.

For each liquidity state, the study calculates the technical-indicator gain by comparing Information Set C with Information Set B on the same stock-date observations. A positive gain means adding the technical block lowers forecast loss. The state-specific results are descriptive; the formal heterogeneity test uses contrasts in the daily cross-sectional mean B-minus-C QLIKE loss differential.

F.2 State-specific locked-test QLIKE results

The nonlinear models show the clearest descriptive liquidity pattern. LightGBM records positive technical-indicator gains in Low and Medium liquidity states at all three horizons, while C deteriorates relative to B in the High liquidity state at every horizon. The pooled LSTM also performs least favourably in the High liquidity state, with deteriorations of 3.13%, 5.17%, and 9.21% at 5, 10, and 20 days. Log-HAR-X is less monotonic: its largest short-horizon gains occur in the medium state, while all three liquidity states deteriorate at 20 days.

Table S27. Locked-test QLIKE by primary turnover-based liquidity state

Model	h	Liquidity state	N	QLIKE B	QLIKE C	TI gain (%)
Log-HAR-X	5	Low	4,946	0.3797	0.3780	0.47
Log-HAR-X	5	Medium	5,665	0.4582	0.4525	1.24
Log-HAR-X	5	High	5,797	0.4361	0.4364	-0.06
Log-HAR-X	10	Low	4,661	0.2650	0.2650	0.01
Log-HAR-X	10	Medium	5,485	0.3244	0.3182	1.91
Log-HAR-X	10	High	5,634	0.2977	0.2961	0.57
Log-HAR-X	20	Low	4,280	0.1756	0.1801	-2.52
Log-HAR-X	20	Medium	5,136	0.2295	0.2303	-0.34
Log-HAR-X	20	High	5,345	0.1933	0.1971	-1.98
LightGBM	5	Low	4,946	0.3821	0.3740	2.13
LightGBM	5	Medium	5,665	0.4511	0.4458	1.17
LightGBM	5	High	5,797	0.4195	0.4269	-1.75
LightGBM	10	Low	4,661	0.2658	0.2638	0.77
LightGBM	10	Medium	5,485	0.3149	0.3027	3.86
LightGBM	10	High	5,634	0.2929	0.2986	-1.92
LightGBM	20	Low	4,280	0.1784	0.1754	1.67
LightGBM	20	Medium	5,136	0.2182	0.2102	3.63
LightGBM	20	High	5,345	0.1960	0.2017	-2.95
LSTM	5	Low	4,946	0.3859	0.3857	0.07
LSTM	5	Medium	5,665	0.4497	0.4568	-1.57
LSTM	5	High	5,797	0.4262	0.4396	-3.13
LSTM	10	Low	4,661	0.2661	0.2679	-0.66
LSTM	10	Medium	5,485	0.3207	0.3235	-0.89

Model	h	Liquidity state	N	QLIKE B	QLIKE C	TI gain (%)
LSTM	10	High	5,634	0.2934	0.3086	-5.17
LSTM	20	Low	4,280	0.1814	0.1825	-0.61
LSTM	20	Medium	5,136	0.2412	0.2499	-3.61
LSTM	20	High	5,345	0.1938	0.2117	-9.21

$Gain = 100 \times [QLIKE(B) - QLIKE(C)] / QLIKE(B)$. Positive values favour technical indicators. The common frozen forecasting panel defines liquidity-state membership, so all three architectures use the same state counts.

F.3 Secondary MSE evidence by liquidity state

MSE does not provide a stable alternative liquidity story. Under Log-HAR-X, for example, the High-liquidity state shows large MSE deteriorations even when the corresponding short-horizon QLIKE changes are small. LightGBM MSE changes are generally close to zero, while LSTM changes vary in sign. The divergence reinforces the study's decision to treat QLIKE as the primary volatility-forecast loss.

Table S28. Locked-test MSE by primary turnover-based liquidity state

Model	h	Liquidity state	MSE B	MSE C	MSE gain (%)
Log-HAR-X	5	Low	0.0000048	0.0000047	2.41
Log-HAR-X	5	Medium	0.0000127	0.0000128	-0.66
Log-HAR-X	5	High	0.0000343	0.0000375	-9.12
Log-HAR-X	10	Low	0.0000023	0.0000022	1.18
Log-HAR-X	10	Medium	0.0000058	0.0000058	0.38
Log-HAR-X	10	High	0.0000083	0.0000086	-3.81
Log-HAR-X	20	Low	0.0000012	0.0000012	-2.03
Log-HAR-X	20	Medium	0.0000030	0.0000030	-0.60
Log-HAR-X	20	High	0.0000030	0.0000033	-9.92
LightGBM	5	Low	0.0000040	0.0000040	-0.17
LightGBM	5	Medium	0.0000106	0.0000106	-0.07
LightGBM	5	High	0.0000142	0.0000143	-0.51
LightGBM	10	Low	0.0000021	0.0000021	-0.21
LightGBM	10	Medium	0.0000056	0.0000055	0.72
LightGBM	10	High	0.0000082	0.0000082	-0.30
LightGBM	20	Low	0.0000012	0.0000012	0.18
LightGBM	20	Medium	0.0000029	0.0000029	0.18
LightGBM	20	High	0.0000030	0.0000030	-1.48
LSTM	5	Low	0.0000044	0.0000044	0.10
LSTM	5	Medium	0.0000111	0.0000110	0.90
LSTM	5	High	0.0000148	0.0000147	0.62
LSTM	10	Low	0.0000023	0.0000023	-1.06
LSTM	10	Medium	0.0000058	0.0000058	0.75
LSTM	10	High	0.0000085	0.0000085	-0.58
LSTM	20	Low	0.0000014	0.0000014	0.87
LSTM	20	Medium	0.0000031	0.0000031	-1.58
LSTM	20	High	0.0000030	0.0000031	-5.19

Positive MSE gain means lower MSE under C than B. MSE is secondary to QLIKE in the study design.

F.4 Formal liquidity-state contrast inference

The study conducts formal inference using daily cross-sectional B-minus-C QLIKE loss differentials. For each date and liquidity state, the analysis averages the stock-level loss differentials across available firms and then forms state contrasts over calendar time. A positive Low-minus-High contrast indicates that technical indicators are relatively more favourable in the Low-liquidity state than in the High-liquidity state. The other ordered contrasts follow the same interpretation.

The study addresses dependence in two ways. First, Newey-West/HAC standard errors use a lag of at least $h-1$ to account for overlapping multi-day targets. Second, the moving-block bootstrap resamples calendar-time blocks while preserving the cross-section within each date. Each bootstrap uses 5,000 replications. The study applies Holm adjustment across the three horizons within each model-contrast family.

Table S29A. Formal liquidity-state contrasts: Log-HAR-X

Contrast	h	Mean contrast	Bootstrap 95% CI	HAC p	Bootstrap p	Holm-HAC p	Holm-bootstrap p
Low – High	5	0.0030	[-0.0083, 0.0141]	0.5739	0.5822	1.0000	1.0000
Low – High	10	-0.0011	[-0.0099, 0.0071]	0.8089	0.8032	1.0000	1.0000
Low – High	20	0.0003	[-0.0058, 0.0059]	0.9261	0.9230	1.0000	1.0000
Medium – High	5	0.0055	[-0.0045, 0.0149]	0.2610	0.2692	0.7829	0.7572
Medium – High	10	0.0045	[-0.0060, 0.0142]	0.3956	0.3772	0.7829	0.7572
Medium – High	20	0.0033	[-0.0027, 0.0085]	0.2621	0.2524	0.7829	0.7572
Low – Medium	5	-0.0025	[-0.0118, 0.0074]	0.5642	0.6058	0.6335	0.6058
Low – Medium	10	-0.0056	[-0.0145, 0.0032]	0.2112	0.1994	0.6335	0.5982
Low – Medium	20	-0.0030	[-0.0088, 0.0028]	0.2996	0.2950	0.6335	0.5982

No Log-HAR-X pairwise liquidity contrast is significant before or after Holm adjustment.

Table S29B. Formal liquidity-state contrasts: LightGBM

Contrast	h	Mean contrast	Bootstrap 95% CI	HAC p	Bootstrap p	Holm-HAC p	Holm-bootstrap p
Low – High	5	0.0155	[0.0019, 0.0308]	0.0241	0.0338	0.0722	0.1014
Low – High	10	0.0082	[-0.0033, 0.0211]	0.1827	0.1826	0.2300	0.2276
Low – High	20	0.0089	[-0.0013, 0.0209]	0.1150	0.1138	0.2300	0.2276
Medium – High	5	0.0119	[-0.0041, 0.0325]	0.1578	0.1922	0.2580	0.2598
Medium – High	10	0.0170	[-0.0003, 0.0419]	0.1197	0.1042	0.2580	0.2598
Medium – High	20	0.0120	[-0.0001, 0.0273]	0.0860	0.0866	0.2580	0.2598
Low – Medium	5	0.0036	[-0.0193, 0.0231]	0.7099	0.7416	1.0000	1.0000
Low – Medium	10	-0.0088	[-0.0336, 0.0091]	0.4293	0.4188	1.0000	1.0000
Low – Medium	20	-0.0031	[-0.0215, 0.0104]	0.7017	0.7098	1.0000	1.0000

The 5-day Low – High contrast is positive and significant before multiplicity adjustment under both HAC ($p = 0.0241$) and bootstrap inference ($p = 0.0338$), but not after Holm adjustment (0.0722 and 0.1014).

Table S29C. Formal liquidity-state contrasts: LSTM

Contrast	h	Mean contrast	Bootstrap 95% CI	HAC p	Bootstrap p	Holm-HAC p	Holm-bootstrap p
Low – High	5	0.0141	[0.0014, 0.0279]	0.0229	0.0374	0.0688	0.1122
Low – High	10	0.0138	[0.0004, 0.0285]	0.0518	0.0526	0.1035	0.1122
Low – High	20	0.0157	[-0.0060, 0.0485]	0.2805	0.2712	0.2805	0.2712
Medium – High	5	0.0064	[-0.0100, 0.0228]	0.3846	0.4530	0.7693	0.9060
Medium – High	10	0.0118	[-0.0054, 0.0287]	0.1756	0.1772	0.5267	0.5316
Medium – High	20	0.0079	[-0.0197, 0.0453]	0.6421	0.6598	0.7693	0.9060
Low – Medium	5	0.0077	[-0.0034, 0.0200]	0.1777	0.1956	0.5330	0.5868
Low – Medium	10	0.0019	[-0.0103, 0.0156]	0.7744	0.7582	0.7744	0.7582
Low – Medium	20	0.0078	[-0.0082, 0.0259]	0.3741	0.3714	0.7482	0.7428

The 5-day Low – High contrast is positive and significant before multiplicity adjustment under both HAC ($p = 0.0229$) and bootstrap inference ($p = 0.0374$), but not after Holm adjustment (0.0688 and 0.1122). The 10-day contrast is borderline even before adjustment.

F.5 What survives multiplicity adjustment?

None of the 27 formal liquidity contrasts is significant at the 5% level under both Holm-adjusted HAC and Holm-adjusted block-bootstrap inference. The strongest directional evidence appears in the 5-day

Low-minus-High contrast for LightGBM and the pooled LSTM. Both are positive and significant under the unadjusted HAC and bootstrap tests, but both fall outside the 5% threshold after adjustment across horizons.

This distinction is central to RQ2. The state-specific loss tables show economically interesting heterogeneity, especially a tendency for the nonlinear models to perform less favourably after adding technical indicators in the most liquid tercile. Formal inference, however, is not strong enough to label the pattern confirmatory. The manuscript therefore describes RQ2 as suggestive rather than established.

F.6 Alternative liquidity and staleness measures in the frozen design

The frozen forecasting panel also contains `zero_ret20`, `log_amihud20` and `obs_share20`. `zero_ret20` is the recent share of zero close-to-close returns; `log_amihud20` is the logged 20-session Amihud price-impact proxy. The study calculates `obs_share20` as the fraction of the preceding 20 NSE market dates on which it observes the stock. The first two variables enter the broader liquidity robustness design, while turnover remains the pre-specified primary state variable.

Importantly, the completed confirmatory/postprocessing archive contains formal state-contrast inference only for the turnover-based states. The Supporting Information therefore avoids introducing new post-hoc state tests for zero-return share, Amihud illiquidity, or observation share after the study freezes the empirical design. Those variables remain available in the replication panel for future extensions or reviewer-requested sensitivity analysis.

Table S30. Liquidity and staleness measures retained in the frozen panel

Measure	Direction	Role in frozen design	State-analysis status
<code>log_turnover20</code>	Higher values = more liquid	Primary state variable	Cross-sectional date-specific terciles
<code>zero_ret20</code>	Higher values = less continuous / less liquid	Information Set B input; robustness measure	No new post-hoc state inference added
<code>log_amihud20</code>	Higher values = more illiquid	Information Set B input; robustness measure	No new post-hoc state inference added
<code>obs_share20</code>	Higher values = greater observation/trading continuity	Staleness/coverage robustness	No new post-hoc state inference added

F.7 Interpretation and scope

The liquidity evidence is consistent with, but does not identify, several possible market-microstructure channels. In less continuously traded securities, technical transformations may summarise gradual price adjustment or persistence that is not readily apparent from individual underlying market observations. Conversely, stale prices, zero returns, and discrete repricing can create apparent technical patterns without stable latent-volatility information. The observational state design cannot distinguish among these competing mechanisms because they imply different predictions.

For that reason, the empirical result should remain phrased in terms of liquidity-conditioned information value. The study does not claim that liquidity makes technical indicators more or less useful. Establishing such a mechanism would require intraday microstructure data, direct measures of spreads/depth, or exogenous variation in liquidity.

GJR-GARCH Benchmark, Fit Diagnostics and Forecast-Positivity Audits

G.1 Benchmark role and specification

The benchmark is a stock-specific GJR-GARCH(1,1) model with an intercept-only conditional mean and standardised Student-t innovations. The study estimates the model on close-to-close log returns multiplied by 100 and recursively updates it each quarter. The asymmetric GJR term allows negative return innovations to affect conditional variance differently from positive innovations, following Glosten, Jagannathan and Runkle (1993).

For each forecast origin, the h-step conditional-variance path is generated recursively and averaged across the next h trading steps before conversion back from percentage-return variance to raw log-return variance. The resulting quantity is therefore comparable in scale with the Yang-Zhang target, which is an average daily variance over the future window. Because the GJR model uses only close-to-close returns while the main models use richer OHLC-derived targets and information sets, benchmark comparisons are descriptive rather than a claim of architectural equivalence. Retaining a conventional GARCH-family benchmark is also consistent with the forecast-comparison role emphasised by Hansen and Lunde (2005).

G.2 GJR estimation success and persistence diagnostics

All 783 scheduled stock-refit estimations complete successfully with convergence code zero. The fit registry contains 493 validation-stage refits and 290 locked-test refits. Estimated persistence is high for many stock periods, which is unsurprising for daily volatility models, but this makes forecast diagnostics particularly important in a thinly traded setting because long runs of small close-to-close returns can drive return-based conditional variance far below an OHLC range-based target.

Table S31. GJR-GARCH fit and persistence audit

Stage	Fits	Status ok	Conv.=0	Median train returns	P10 persistence	Median persistence	P90 persistence	Persistence $\geq .999$
Locked test	290	290	290	2152	0.839	0.999	0.999	2
Validation	493	493	493	1342	0.769	0.999	0.999	1

Persistence is $\alpha + \beta + \gamma/2$ from the fitted GJR specification. Values reported as 0.999 are fitted values recorded by the completed estimation pipeline, not post-hoc replacements introduced for this SI.

G.3 Pooled and equal-stock benchmark performance

The locked-test benchmark is economically interpretable but clearly weaker than the information-rich models. Pooled GJR QLIKE equals 0.9263, 0.7485 and 0.7402 at 5, 10 and 20 days. Equal-stock-weight mean QLIKE is very similar at 5 and 10 days and somewhat lower at 20 days. By contrast, validation-stage mean QLIKE is dominated by one thin-trading case, so median-stock losses better represent the typical validation firm.

Table S32. GJR-GARCH pooled and equal-stock-weight performance

Stage	h	Forecasts	Pooled QLIKE	Mean stock QLIKE	Median stock QLIKE	P90 stock QLIKE	Pooled MSE
Locked test	5	15,899	0.9263	0.9212	0.7295	1.0939	1.30×10^{-5}
Locked test	10	15,313	0.7485	0.7388	0.4466	0.7556	8.61×10^{-6}
Locked test	20	14,278	0.7402	0.7111	0.4000	0.6092	6.59×10^{-6}
Validation	5	27,968	1.025e+06	9.962e+05	0.7012	1.1552	7.19×10^{-6}
Validation	10	26,611	9.4e+05	9.167e+05	0.4397	0.9604	6.49×10^{-6}
Validation	20	24,792	6.158e+05	6.186e+05	0.3637	0.8957	7.76×10^{-6}

The extreme validation QLIKE values arise from a near-zero GJR-GARCH forecast episode affecting a small number of stock-period observations. Because QLIKE strongly penalises forecasts that approach zero when realised variance remains positive, these observations dominate the pooled and mean-stock losses. The median-stock QLIKE values show that the extreme losses are not representative of the typical stock.

G.4 Common-support comparison with Log-HAR-X Information Set C

On exact common stock-date support, Log-HAR-X with Information Set C substantially outperforms the conventional return-based GJR benchmark in the locked sample. The QLIKE reduction is 55.78% at five days, 60.84% at 10 days, and 71.99% at 20 days. These comparisons provide context for the value of richer volatility memory and market information, but they are not the paper's central estimand.

Table S33. Log-HAR-X C versus GJR-GARCH on exact common support

Stage	h	N	QLIKE Log-HAR-X C	QLIKE GJR	Log-HAR-X C gain (%)
Validation	5	25,427	0.4331	4.2×10^4	100.00
Validation	10	24,200	0.3065	5.246×10^4	100.00
Validation	20	22,527	0.2469	6.346×10^4	100.00
Locked test	5	14,848	0.4138	0.9358	55.78
Locked test	10	14,296	0.2923	0.7465	60.84
Locked test	20	13,339	0.2033	0.7258	71.99

$Gain = 100 \times [QLIKE(GJR) - QLIKE(Log-HAR-X C)] / QLIKE(GJR)$. The near-zero GJR forecast episode documented in Section G.5 drives the extreme validation gains, so the analysis does not interpret them as a general percentage advantage.

G.5 Nairobi Business Ventures thin-trading diagnostic

Nairobi Business Ventures (NBV) explains all of the explosive validation-stage GJR QLIKE. During validation, some NBV GJR variance forecasts fall below 10^{-12} while the OHLC-based Yang-Zhang target remains positive. Because QLIKE penalises severe variance underprediction sharply, NBV contributes more than 99.9999% of aggregate validation QLIKE at every horizon. Excluding NBV reduces pooled validation QLIKE from values above 600,000-1,000,000 to approximately 0.44-0.74. The diagnostic relevance of thin trading is consistent with earlier African-market evidence that non-trading can materially affect return-based inference (Appiah-Kusi & Menyah, 2003).

The diagnostic does not justify deleting NBV from the study. The firm satisfies the ex-ante eligibility rule, and removing it because a conventional benchmark performs poorly would introduce outcome-based sample selection. Instead, the paper retains NBV, reports pooled and robust stock-level summaries, and treats the episode as evidence that close-to-close return-based volatility models can behave pathologically when observed returns are extremely stale relative to the intraday OHLC range.

Table S34. Influence of Nairobi Business Ventures on pooled GJR QLIKE

Stage	h	Pooled QLIKE	QLIKE excl. NBV	NBV mean QLIKE	NBV share total QLIKE (%)	NBV forecasts
Locked test	5	0.9263	0.7074	6.846	26.35837	567
Locked test	10	0.7485	0.4844	7.745	37.63832	557
Locked test	20	0.7402	0.3979	9.499	48.26240	537
Validation	5	1.025×10^6	0.7395	2.889×10^7	99.99993	992
Validation	10	9.4×10^5	0.5035	2.658×10^7	99.99995	941
Validation	20	6.158×10^5	0.4408	1.794×10^7	99.99993	851

Table S35. NBV GJR forecast-path diagnostics

Stage	h	N	Mean target	Mean forecast	Min forecast	Forecast < $1e-12$ (%)	QLIKE
Locked test	5	567	2.10e-03	1.14e-03	2.96e-06	0.00	6.846
Locked test	10	557	2.08e-03	1.13e-03	2.95e-06	0.00	7.745

Stage	h	N	Mean target	Mean forecast	Min forecast	Forecast <1e-12 (%)	QLIKE
Locked test	20	537	2.05e-03	1.11e-03	2.92e-06	0.00	9.499
Validation	5	992	2.85e-03	2.09e-03	1.68e-19	4.13	2.889×10^7
Validation	10	941	2.84e-03	2.06e-03	3.50e-19	3.08	2.658×10^7
Validation	20	851	2.83e-03	2.06e-03	6.50e-19	2.12	1.794×10^7

By the locked test, NBV no longer produces forecasts below 10^{-12} , but it remains the highest-QLIKE GJR stock at each horizon. The persistent stock-level disparity motivates reporting pooled, equal-stock-weighted, and stock-level summaries rather than relying on pooled performance alone.

G.6 Complete locked-test stock-level GJR QLIKE

The full stock cross-section shows that the benchmark problem is concentrated rather than universal. Most firms have locked-test stock-level QLIKE below one at the 10- and 20-day horizons, while NBV is a clear outlier. Reporting all 29 firms prevents the pooled benchmark result from being mistaken for uniform model failure.

Table S36. Complete locked-test stock-level GJR-GARCH diagnostics

Firm	N at 5d	QLIKE 5d	QLIKE 10d	QLIKE 20d	Minimum forecast across horizons
Absa Bank Kenya	573	0.799	0.442	0.230	0.000205
Britam Holdings	567	1.102	0.799	0.464	0.000282
CIC Insurance Group	526	0.919	0.745	0.586	0.000570
Carbacid Investments Ltd	543	0.496	0.341	0.318	0.000468
Centum	561	0.897	0.431	0.139	0.000363
Co-operative Bank (Kenya)	555	0.482	0.257	0.269	0.000400
Diamond Trust Bank	573	0.682	0.669	0.817	0.000602
East African Breweries	555	0.744	0.357	0.235	0.000365
Equity Group	567	0.381	0.278	0.400	0.000367
Eveready	561	0.618	0.526	0.473	0.001100
Flame Tree Group	504	1.146	1.045	0.408	0.000613
HF Group	561	0.562	0.330	0.194	0.000738
Home Africa	549	0.739	0.655	0.611	0.001150
JUBILEE HOLDINGS	531	1.012	0.631	0.437	0.000207
KCB Bank	557	0.677	0.395	0.352	0.000208
KPLC	567	0.602	0.437	0.482	0.000729
KenGen	567	0.679	0.502	0.539	0.000442
Kenya Re	567	1.092	0.640	0.346	0.000326
Longhorn Publishers	426	0.580	0.428	0.342	0.000450
NCBA Bank	567	0.380	0.406	0.609	0.000614
Nairobi Business Ventures	567	6.846	7.745	9.499	0.000003
Nairobi Securities Exchange	549	0.361	0.165	0.162	0.000549
Nation Media Group	527	0.938	0.509	0.200	0.000409
Safaricom PLC	561	0.338	0.261	0.323	0.000310
Sasini	527	0.855	0.628	0.521	0.000796
Standard Chartered Bank	561	0.848	0.447	0.293	0.000180
Total Energy	510	0.979	0.642	0.506	0.000757
Uchumi Supermarkets	573	0.730	0.540	0.481	0.001170
WWP Scangroup	547	0.228	0.175	0.384	0.001870

N differs modestly by horizon because longer Yang-Zhang targets are less frequently available. The minimum forecast column is the lowest positive GJR forecast observed for the firm across the three locked horizons.

G.7 Forecast-positivity and fit-success audits across architectures

The final audit distinguishes model-estimation success from forecast-path quality. Log-HAR-X, LightGBM and the pooled LSTM produce no non-positive forecasts in the recorded audit, and Log-HAR-X has no numerical-floor hits. GJR forecasts are also strictly positive, but positivity alone is not sufficient: the NBV validation episode demonstrates that extremely small positive variance forecasts can still generate severe QLIKE loss.

Table S37. Model-wide fit-success and positivity audit

Architecture	Recorded fits	Successful fits	Forecasts audited	Non-positive	Minimum forecast	Safeguard/interpretation
Log-HAR-X	630	630	749,274	0	0.000002	Exponentiation; no numerical-floor hits
LightGBM	54	54	282,312	0	0.000021	Log-target retransformation
LSTM	54	54	282,312	0	0.000225	Log-target retransformation
GJR-GARCH	783	783	124,861	0	1.68E-19	Positive conditional-variance recursion; near-zero values occur in validation

Table S38. GJR forecast-positivity and near-zero audit

Stage	h	Forecasts	Minimum	0.1th percentile	Non-positive	<1e-12	Share <1e-12 (%)
Locked test	5	15,899	0.0000030	0.0000313	0	0	0.0000
Locked test	10	15,313	0.0000030	0.0000301	0	0	0.0000
Locked test	20	14,278	0.0000029	0.0000291	0	0	0.0000
Validation	5	27,968	1.68E-19	3.63E-15	0	41	0.1466
Validation	10	26,611	3.50E-19	7.56E-13	0	29	0.1090
Validation	20	24,792	6.50E-19	3.59E-11	0	18	0.0726

Near-zero GJR forecasts occur only during validation, with NBV accounting for most of these cases. The study records no non-positive GJR forecasts in either the validation or locked-test stage.

G.8 Interpretation for the manuscript

The GJR benchmark serves two purposes. First, it confirms that the information-rich forecasting systems materially improve on a conventional asymmetric return-variance benchmark in the locked sample. Second, its validation pathology illustrates why benchmark diagnostics matter in a thinly traded frontier market. The correct response is not to delete the difficult stock, but to disclose the loss concentration and supplement pooled statistics with stock-level and equal-stock summaries.

For the central research question, however, the GJR results remain secondary. The study addresses RQ1 through the nested B-versus-C comparison within each main architecture while keeping the model structure, refit schedule, and tuning fixed. The GJR model therefore belongs primarily in Supporting Information rather than in the headline technical-indicator evidence.

References

- Amihud, Y. (2002) 'Illiquidity and stock returns: cross-section and time-series effects', *Journal of Financial Markets*, Vol. 5, No. 1, pp.31–56, [https://doi.org/10.1016/S1386-4181\(01\)00024-6](https://doi.org/10.1016/S1386-4181(01)00024-6).
- Appiah-Kusi, J. and Menyah, K. (2003) 'Return predictability in African stock markets', *Review of Financial Economics*, Vol. 12, No. 3, pp.247–270, [https://doi.org/10.1016/S1058-3300\(02\)00073-3](https://doi.org/10.1016/S1058-3300(02)00073-3).
- Bollinger, J. (2001) *Bollinger on Bollinger Bands*, McGraw-Hill.
- Glosten, L.R., Jagannathan, R. and Runkle, D.E. (1993) 'On the relation between the expected value and the volatility of the nominal excess return on stocks', *The Journal of Finance*, Vol. 48, No. 5, pp.1779–1801, <https://doi.org/10.1111/j.1540-6261.1993.tb05128.x>.
- Hansen, P.R. and Lunde, A. (2005) 'A forecast comparison of volatility models: does anything beat a GARCH(1,1)?', *Journal of Applied Econometrics*, Vol. 20, No. 7, pp.873–889, <https://doi.org/10.1002/jae.800>.
- Wilder, J.W. Jr. (1978) *New Concepts in Technical Trading Systems*, Trend Research.
- Yang, D. and Zhang, Q. (2000) 'Drift-independent volatility estimation based on high, low, open, and close prices', *The Journal of Business*, Vol. 73, No. 3, pp.477–492, <https://doi.org/10.1086/209650>.